Research Report

2026 S.T. Yau High School Science Award (Asia)

Bundle Neural Networks for Functional Connectomes

Density-Dependent Aggregation and Cross-Site Prediction

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Report Version	Completed evidence report with E1 mechanism audit, 14 August 2026

Completed evidence draft. This version reports the frozen ABIDE-I operator study and a separately labelled, audited post-hoc intervention analysis of its accepted BuNN checkpoints. External validation and additional architecture studies are not presented as completed results.

This document is a separate report-format rendering. It does not replace the existing IEEE-style manuscript. Before any actual submission, the researcher must independently revise the prose, complete the acknowledgments, verify all personal details, and sign the integrity commitments.

Abstract

Functional connectomes have a natural graph representation, but the usual node feature is already a complete connectivity profile. Neighborhood aggregation may therefore blur useful differences instead of supplying new information. This study tested whether learned orthogonal bundle transport changes that density-dependent behavior. The analysis used 754 technically eligible ABIDE-I participants from 18 sites. A shared neural backbone was paired with identity propagation, a learned-local capacity control, Graph Convolutional Network (GCN) aggregation, trivial-bundle diffusion, or learned Bundle Neural Network (BuNN) transport. Nonzero graphs retained the strongest positive edges at 1, 5, 10, and 20 percent density. Evaluation used nested held-out-site validation, five final seeds, regularized non-graph baselines, and common-frame representation diagnostics.

The primary BuNN-minus-GCN performance-curve contrast was -0.00958 (95% site-bootstrap confidence interval [-0.03665, 0.01146]; exact sign-flip p = 0.524). Relative to the connectome elastic net, the BuNN contrast was -0.05516 (95% confidence interval [-0.08297, -0.02752]). The matched-anchor effective-rank contrast was -0.00719 (95% confidence interval [-0.01310, -0.00105]), which ran against the proposed representation-preservation advantage. Performance declined as graph density increased for each diffusion operator. Site weighting and training seed also affected the small BuNN–GCN difference.

A separately frozen post-hoc intervention analysis then tested what the 360 accepted BuNN checkpoints actually used. Replacing learned maps with random orthogonal maps reduced equal-site balanced accuracy by 8.86 percentage points (95% interval $[-12.36, -4.78]$; Holm-adjusted exact $p = 0.0038$). Identity maps, shuffled learned maps, and degree-preserving graph rewiring caused smaller reductions and did not pass the four-contrast multiplicity-controlled test. Representation changes were substantial but only weakly associated with prediction changes across site-density cells.

Under this specified ABIDE-I pipeline, learned bundle transport provided no detected advantage over GCN or the elastic net. The intervention audit shows that its learned maps nevertheless influenced its own predictions; the exact graph topology contributed much less. The study does not establish a biological bundle geometry, a clinical diagnostic system, or a universal result for BuNNs. Its contribution is a controlled operator comparison and a checkpoint-level audit in a dense, globally informed connectome setting.

Keywords: functional connectome; bundle neural network; graph neural network; representation collapse; held-out-site validation; ABIDE-I; autism

Acknowledgments and Assistance Statement

The study used public, anonymized ABIDE-I derivatives distributed through the Preprocessed Connectomes Project. The original ABIDE investigators and participating institutions collected and shared those data; they were not involved in the present analysis. The computational work used open-source Python, PyTorch, NumPy, pandas, scikit-learn, and scientific plotting tools, with training performed on an AWS EC2 instance equipped with an NVIDIA A10G GPU.

Supervisor contribution. The researcher must replace this sentence with an exact account of the supervising teacher's intellectual, technical, editorial, and logistical contributions. No contribution should be inferred or invented.

Research Integrity Commitments

The following page is included for format readiness. It is not signed in this completed evidence draft.

1. The reported data, code outputs, analyses, and citations must be honest, traceable, and presented without selective omission.
2. The student researcher must understand and be able to explain the methods, implementation, results, limitations, and conclusions.
3. All external assistance must be disclosed in accordance with the applicable competition rules.
4. Public data and software must be credited, and participant-level data must not be exposed in the report or release package.
5. No incomplete, exploratory, or planned experiment may be described as a completed result.
6. Any relationship that could create a conflict of interest with judging or supervision must be disclosed.
7. The work must comply with relevant safety, research-integrity, and competition requirements.
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Contents

1 Introduction

Resting-state functional magnetic resonance imaging records slow changes in the blood-oxygen-level-dependent signal while a participant is not performing a specific task. A functional connectome summarizes the statistical association between pairs of regional time series. Brain regions become graph nodes, and correlations between their signals become weighted edges. This representation makes graph neural networks an attractive modeling choice.

That attraction contains a hidden assumption: message passing must add useful information. In many connectome classifiers, each node already receives one full row of the correlation matrix. Its feature vector therefore describes its association with every region in the atlas. Aggregating neighboring rows does not introduce context that was previously absent. It may instead average away distinctions that a classifier needs.

Recent controlled benchmarking found that classical models and non-graph neural networks can match or outperform graph deep-learning models on several functional-connectome tasks. Ablations in that work implicated message aggregation as one source of degradation [1]. A graph representation alone therefore does not justify graph propagation. The operator must be compared with no propagation and with strong regularized baselines on the same data splits.

BuNNs offer a distinct propagation rule. Before diffusion, they use learned orthogonal maps to transport node features into compatible coordinate frames [2]. Theoretical and synthetic results suggest that this construction can resist representation collapse more effectively than ordinary diffusion. Neural-sheaf methods provide a related example of how non-trivial transport can alter message passing [3]. The open question is whether this advantage carries into dense functional connectomes with globally informed node features.

1.1 How the original question was refined

The first formulation of this project treated negative functional correlations as a possible form of heterophilic message passing. Formalizing the experiment showed that this interpretation was not supportable. Correlation sign does not identify graph-learning heterophily, anatomical connections, neural excitation or inhibition, or causal flow. It can also change with preprocessing choices such as global-signal regression [4]. The original dataset, connectome representation, BuNN architecture, and aggregation problem were retained, but the hypothesis was narrowed to the computational behavior of the propagation operator.

The resulting study asks:

> On a pre-specified ABIDE-I pipeline, does learned bundle transport produce a more favorable held-out-site performance–density curve than GCN aggregation, and does any predictive pattern co-occur with stronger common-frame representation preservation?

An advantage was required to exceed GCN, matched no-propagation controls, and the strongest regularized connectome baseline. This stricter rule prevents a complex operator from being credited merely because it is less harmful than another graph operator.

2 Background

2.1 ABIDE-I and cross-site prediction

ABIDE-I combined resting-state fMRI and phenotypic data from international sites to support large-scale autism research [5]. The Preprocessed Connectomes Project later distributed derivatives produced by several pipelines, including C-PAC [6]. Multi-site prediction is difficult because scanners, acquisition protocols, participant composition, motion, and preprocessing can differ across sites. A random participant split can place nearly identical site conditions in training and test data. Holding out an entire site creates a stricter evaluation: the model must generalize to a data-collection environment it did not see during fitting.

2.2 Graph convolution and aggregation

A GCN combines a node's features with normalized features from neighboring nodes and then applies a learned transformation [7]. As propagation repeats or the graph becomes denser, representations can contract. Whether this is harmful depends on the graph, input features, architecture, and task. Here the question is especially relevant because each node begins with a complete connectivity row. A no-propagation control is therefore part of the scientific comparison, not merely an ablation.

2.3 Bundle transport and coordinate frames

BuNNs associate node features with local coordinate frames. Learned orthogonal maps align features before diffusion [2]. This creates an important measurement issue. Raw embeddings at different nodes may look dissimilar simply because they are expressed in different frames. Directly stacking them and computing cosine similarity or effective rank could reward rotations instead of retained task information. The present study maps embeddings into a common learned frame before comparing nodes and also uses a learned-local control that retains map-generating capacity without inter-node diffusion.

3 Methods

3.1 Study design and frozen evidence boundary

The experiment isolated the propagation operator while holding the rest of the pipeline fixed. Classical baselines were completed and audited first. Five neural variants then used the same inputs, outer test sites, grouped inner folds, candidate budget, stopping rule, and final seeds. Before results were opened, the complete neural artifact was checked for expected coverage and the confirmatory analysis was frozen. The accepted run contained 9,324 neural fits, 52,780 held-out predictions, and results for all 18 outer sites.

This report uses only the frozen Study 1 evidence. Later mechanism audits, deeper networks, Wang-style time-series models, and ABIDE-II evaluation remain separate extensions and are not included as findings.

3.2 Dataset and technical quality control

The source was the ABIDE-I C-PAC band-pass-filtered AAL ROI time-series derivative without global-signal regression. The AAL atlas divides the brain into 116 macroscopic regions in standardized space [8]. Eligibility required a valid file header, finite numeric values, nonzero variance in every ROI time series, and a finite resulting connectome. Fifteen of 769 downloaded files contained at least one zero-variance ROI, making the correlation undefined. Their exclusion left 754 technically eligible participants: 371 labeled ASD and 383 controls across 18 sites.

Table 1: Frozen cohort and feature summary.

Quantity	Value
Downloaded ABIDE-I ROI time-series files	769
Zero-variance-ROI technical exclusions	15
Technically eligible participants	754
ASD / control participants	371 / 383
Evaluable acquisition sites	18
AAL regions per participant	116
Unique undirected connectome features	6,670

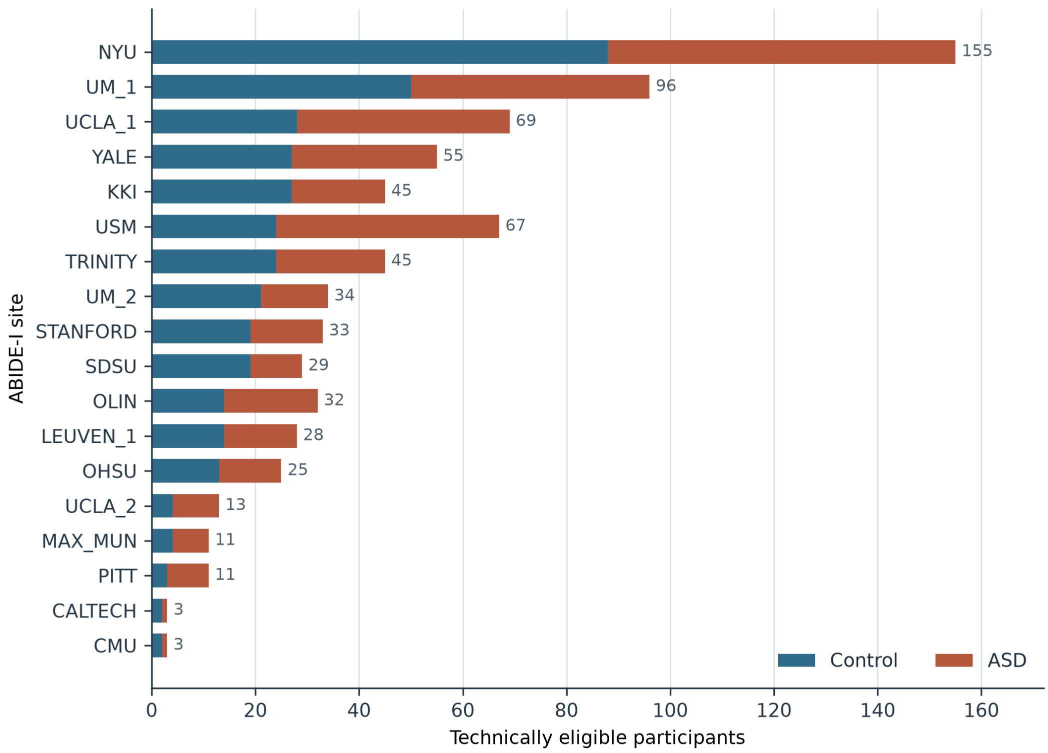

Figure 1: Participant composition of the frozen technically eligible cohort. Sites varied in both total size and class balance, motivating site-level evaluation and equal-site aggregation. This is a descriptive count figure, not a performance result.

Technical eligibility does not imply population representativeness. Site composition remained heterogeneous, as shown in figure 1. That variation is handled in evaluation and uncertainty estimation rather than erased from the data.

3.3 Connectome construction

For each participant, Pearson correlations were computed between every pair of ROI time series and transformed with Fisher's r-to-z function. The diagonal was set to zero. The resulting matrix was finite and symmetric. Its 6,670 lower-triangle entries formed the feature vector for classical models. For neural models, each node received its full 116-value connectivity row.

At nonzero density, graphs retained the strongest positive off-diagonal associations symmetrically at 1, 5, 10, and 20 percent. Density zero supplied the matched identity or learned-local anchor. Negative associations remained part of the node connectivity profiles but were not used as propagation edges. This was a pre-specified computational choice, not a biological claim.

3.4 Classical baselines

Three logistic models provided practical references. The first used age, sex, mean framewise displacement, and usable scan length. The second used all vectorized connectome features with

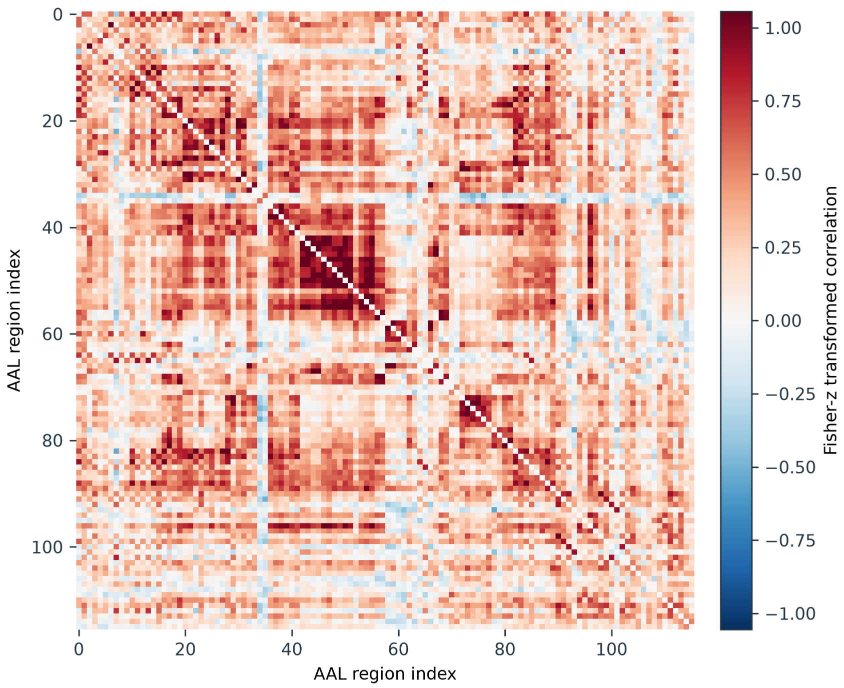

Figure 2: A deterministic example from the frozen archive illustrating a participant-level Fisher-z functional-connectivity matrix. Red and blue indicate positive and negative statistical associations. They are not interpreted as excitation, inhibition, anatomy, or causal flow.

elastic-net regularization, which combines ℓ_1 and ℓ_2 penalties for correlated high-dimensional predictors [9]. The third combined connectome features and covariates. All scaling, feature processing, and tuning were fitted on training sites only.

3.5 Shared neural backbone and operator controls

The neural variants shared the encoder, hidden width, two propagation layers, nonlinearities, normalization, dropout, pooling, classifier, optimizer, training schedule, and tuning budget. The controlled difference was the propagation rule or the map capacity needed to isolate it.

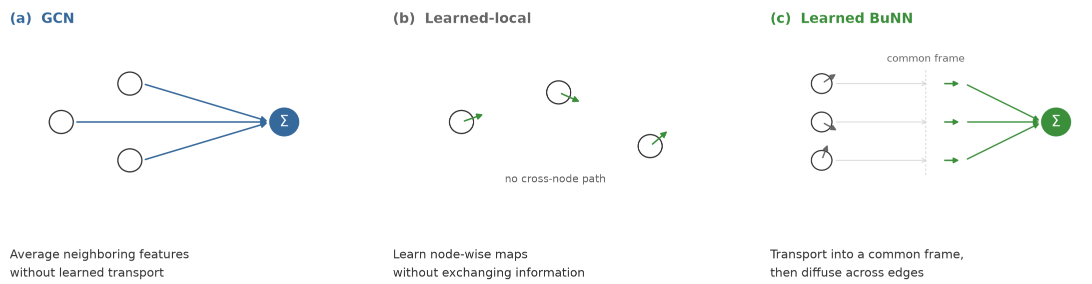

Figure 3: Computational distinction among GCN aggregation, the learned-local capacity control, and learned bundle transport. The schematic does not represent biological signal flow.

Table 2: Neural operator and capacity controls.

Variant	Operation	Purpose
Identity	No inter-node propagation	Tests whether message passing adds value.
Learned-local	Learns the BuNN node-wise maps but does not diffuse between nodes	Separates map-generator capacity from transport.
GCN	Normalized neighborhood aggregation	Standard graph-diffusion reference.
Trivial bundle	Bundle diffusion with identity transport maps	Separates the diffusion construction from learned transport.
Learned BuNN	Learned orthogonal node maps followed by bundle diffusion	Tests the proposed transport operator.

3.6 Nested held-out-site evaluation

Each site served once as the outer test site. Four grouped folds within the remaining sites selected among four equal-budget AdamW candidates using two tuning seeds. The selected configuration was refitted with five fixed final seeds. Test-site data did not enter scaling, tuning, stopping, or model selection. The primary predictive endpoint was the unweighted mean of held-out-site balanced accuracies. This gives each acquisition site equal influence. Pooled balanced accuracy and AUROC were secondary.

3.7 Density-curve and representation estimands

The primary predictive estimand was the normalized trapezoidal area under the balanced-accuracy curve across the full density grid. GCN and trivial-bundle curves used identity at zero density. The learned-BuNN curve used the parameter-matched learned-local anchor. Comparing the whole curve avoided choosing a favorable density after viewing the results.

After each propagation layer, BuNN embeddings were transported into a common learned frame before node comparisons. Normalized effective rank at layer 2 was the primary representation endpoint. Secondary measures included normalized dispersion, mean pairwise cosine similarity, invariant edge-transport distance, and encoder-to-layer-2 rank change. These measures describe co-occurring representation behavior. They do not prove that a representation change caused a prediction change.

3.8 Statistical analysis and decision rule

The five final seeds were averaged within each site and configuration before site-level inference. Confidence intervals used 10,000 paired bootstrap resamples of the 18 held-out sites. Curve contrasts also received exact two-sided paired sign-flip tests. Density-specific tests were secondary and Holm-corrected within each named family.

A positive architectural result required all three of the following:

1. stronger common-frame representation preservation than GCN;
2. better held-out prediction than GCN and the matched no-propagation controls; and
3. performance above the strongest regularized non-graph baseline.

A confidence interval containing zero was reported as no detected advantage. A practically meaningful gain was ruled out only if the upper confidence bound fell below the pre-specified 0.03 balanced-accuracy margin.

3.9 Run integrity and reproducibility

Long-running jobs wrote atomic per-site checkpoints and recorded immutable hashes for data, configuration, and source code. Site ownership, restart validation, canonical merging, and independent completion audits protected the parallel run. The neural artifact was checked for coverage, recomputed metrics, diagnostics, runtime records, and warnings before unblinding. Values in the report are generated from a hash-bound result snapshot rather than copied manually.

3.10 Post-hoc accepted-checkpoint intervention audit

After Study 1 was frozen, a separate protocol tested what the 360 accepted learned-BuNN checkpoints depended on without refitting them. Each checkpoint was first required to reproduce its archived held-out probabilities within 10^{-6}. Four scientific interventions were then compared with unaltered inference: replacing learned maps with identity maps, shuffling learned maps between nodes, replacing them with random orthogonal maps, and rewiring each participant graph by double-edge swaps that preserved its exact degree sequence. An encoded-node permutation served only as an equivariance check.

The randomized families used 100 stored randomizations, shared across the five model seeds. Balanced accuracy was calculated for each seed and randomization, randomizations were averaged within the site-density cell, and seeds were then averaged. The four density-specific paired effects were averaged within site, followed by an unweighted mean over the 18 sites. The effect direction was always intervention minus unaltered BuNN. Uncertainty used 10,000 paired site-bootstrap samples. Two-sided exact sign-flip tests enumerated all 2^{18} site sign assignments, with Holm correction across the four primary interventions.

Secondary endpoints were absolute probability change, classification flips, AUROC, sensitivity, specificity, and changes in the four final-layer gauge-aware diagnostics. Associations between representation and prediction changes were descriptive Spearman correlations across the 72 site-density cells, not mediation tests. The complete run was archived and audited before the analysis protocol and code were frozen. The analyzer passed synthetic known-answer tests before a single acknowledged unblinding invocation, and an independent recomputation audit passed before interpretation.

4 Results

4.1 Classical baseline performance

The connectome elastic net was the strongest classical model, with equal-site balanced accuracy 0.6401. Adding covariates produced 0.6336, while covariates alone produced 0.5652. Imaging features therefore carried cross-site predictive information under this pipeline, but the result is computational and should not be described as clinical diagnostic performance.

Table 3: Held-out-site performance of the classical baselines.

Model	Equal-site BA	Pooled BA	Pooled AUROC
Covariates-only logistic regression	0.5652	0.5629	0.5973
Connectome elastic net	**0.6401**	**0.6495**	**0.6794**
Connectome + covariates elastic net	0.6336	0.6428	0.6773

4.2 Predictive performance across graph density

All diffusion curves declined as graph density increased (figure 4). The learned-BuNN minus GCN normalized curve-area difference was -0.00958 (95% confidence interval [-0.03665, 0.01146]; exact $p = 0.524$). The interval crossed zero, so no predictive advantage was detected. Its upper bound was also below the pre-specified 0.03 margin. This rules out a gain of that size for the specified estimand and pipeline, but not every smaller effect or every other BuNN design.

Against the site-matched connectome elastic net, the learned-BuNN curve difference was -

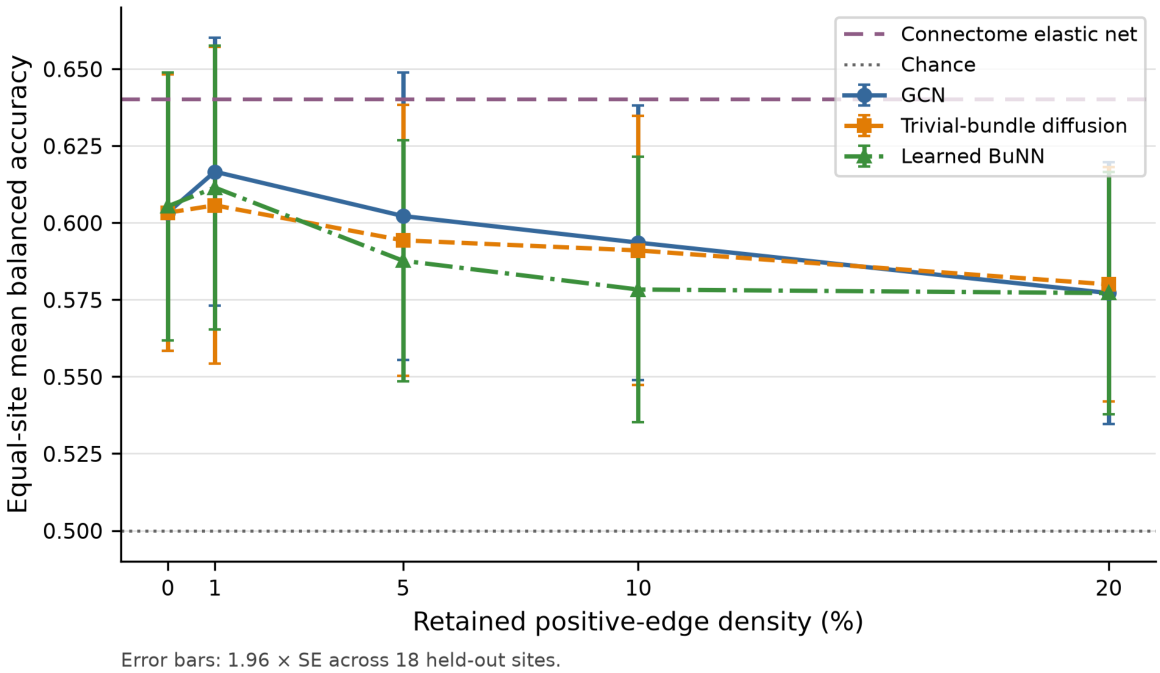

Figure 4: Equal-site mean held-out balanced accuracy across the pre-specified positive-edge density grid. Error bars show 1.96 standard errors across sites for visualization. Confirmatory inference used paired site bootstrap and exact sign-flip tests on the locked curve estimand.

0.05516 (95% confidence interval [-0.08297, -0.02752]; exact $p = 0.0018$). The interval lay entirely below zero. Comparisons with identity and learned-local controls also failed the positive-result rule.

Table 4: Pre-specified predictive contrasts. Estimates are balanced-accuracy differences.

Contrast	Estimate	95% CI	Exact p
BuNN curve − GCN curve	−0.0096	[−0.0366, 0.0115]	0.524
BuNN curve − trivial-bundle curve	−0.0062	[−0.0293, 0.0108]	0.681
BuNN nonzero mean − learned-local	−0.0167	[−0.0347, 0.0009]	0.096
BuNN nonzero mean − identity	−0.0146	[−0.0320, 0.0018]	0.116
BuNN curve − connectome elastic net	−0.0552	[−0.0830, −0.0275]	0.0018

4.3 Common-frame representation behavior

The raw BuNN effective-rank curve appeared higher than the GCN curve, but its learned-local zero-density anchor was already high. After subtracting each operator's matched anchor, the normalized effective-rank contrast was -0.00719 (95% confidence interval [-0.01310, -0.00105]). Its direction was opposite to the proposed BuNN preservation advantage. Normalized dispersion and pairwise cosine similarity showed the same broad pattern.

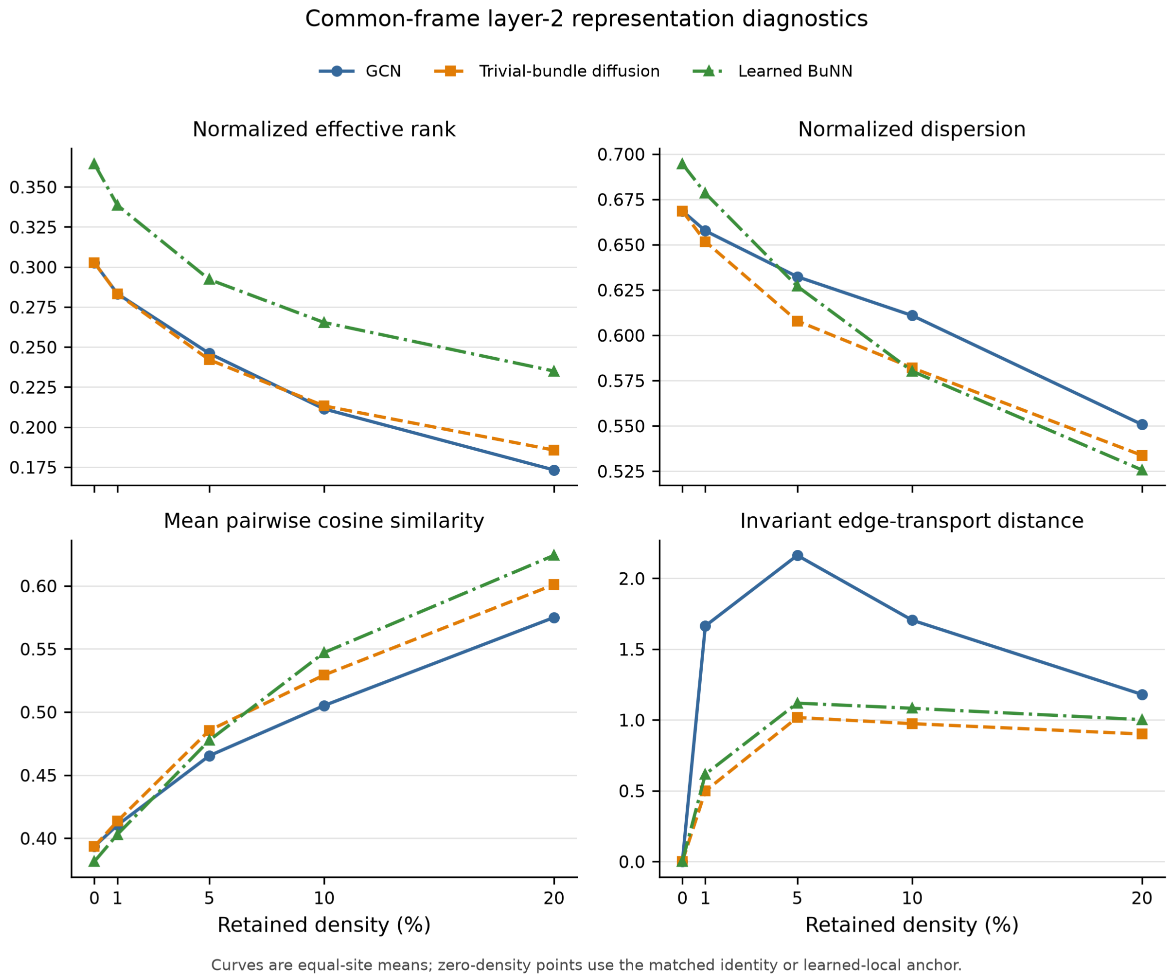

Figure 5: Equal-site common-frame representation diagnostics at layer 2. Zero-density points use the matched identity or learned-local anchor. Raw, unaligned BuNN node embeddings were not compared.

The learned-local control is decisive for interpretation. Without it, extra map-generator capacity could be mistaken for information preserved by inter-node transport. The matched result gives no support for that transport explanation. These diagnostics remain computational measurements and do not identify a biological mechanism.

Table 5: BuNN-minus-GCN matched-anchor representation contrasts.

Layer-2 endpoint	Estimate	95% CI
Normalized effective rank (primary)	−0.0072	[−0.0131, −0.0010]
Normalized dispersion	−0.0417	[−0.0523, −0.0306]
Mean pairwise cosine similarity	0.0415	[0.0319, 0.0507]
Invariant edge-transport distance	−0.6436	[−0.8205, −0.4720]
Encoder-to-layer-2 effective-rank change	−0.0056	[−0.0144, 0.0048]

4.4 Accepted-checkpoint mechanism audit

The post-hoc intervention audit showed that the learned maps were not dormant components. Replacing them with random orthogonal maps reduced held-out balanced accuracy by 8.86 percentage points (95% paired site-bootstrap interval $[-12.36, -4.78]$; exact sign-flip $p = 0.00095$; Holm-adjusted $p = 0.0038$). The effect was negative at every density and in 15 of 18 sites after averaging densities. This was the only intervention that passed the pre-specified four-contrast multiplicity-controlled test.

Identity maps produced a 4.72-point reduction (95% interval $[-8.56, -0.29]$). Its unadjusted exact p was 0.0468, but its adjusted p was 0.1404. Shuffling learned maps between nodes caused a smaller 1.94-point reduction (95% interval $[-3.77, 0.05]$; adjusted $p = 0.1404$). Degree-preserving rewiring caused a 1.23-point reduction (95% interval $[-2.49, -0.05]$; adjusted $p = 0.1404$). Bootstrap intervals and sign-flip tests answer related but different questions; the adjusted sign-flip result controls the four named primary tests.

Table 6: Accepted-checkpoint intervention effects. Estimates are intervened minus unaltered BuNN balanced accuracy in percentage points.

Intervention	Estimate	95% CI	Holm p
Identity maps	−4.72	$[-8.56, -0.29]$	0.1404
Shuffled learned maps	−1.94	$[-3.77, 0.05]$	0.1404
Random orthogonal maps	−8.86	$[-12.36, -4.78]$	**0.0038**
Degree-preserving rewiring	−1.23	$[-2.49, -0.05]$	0.1404

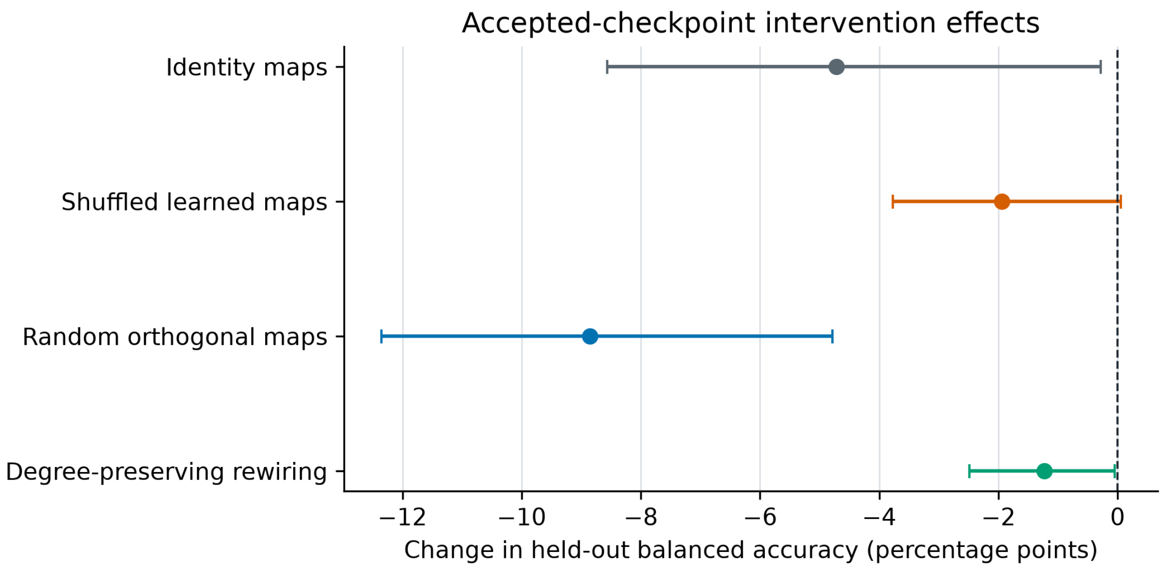

Figure 6: Overall E1 intervention effects with 95% paired site-bootstrap intervals. Negative values mean that disrupting the component reduced the accepted checkpoint's held-out balanced accuracy. Statistical decisions use the Holm-adjusted exact sign-flip tests in table 6.

The random-map effect remained negative at 1, 5, 10, and 20 percent density (figure 7). Identity maps had their largest effect at 1 percent density. Shuffling maps and rewiring topology produced smaller effects that often had density-specific intervals crossing zero. The density pattern therefore does not look like a benefit confined to one hand-picked threshold.

The prediction changes were also practically visible. Random maps changed 49.46% of binary decisions and reduced AUROC by 14.59 points. Identity maps changed 34.41% of decisions and reduced AUROC by 9.50 points. Shuffled maps changed 15.12% of decisions, while topology rewiring changed 6.96%. Random maps strongly reduced sensitivity while increasing specificity, indicating a shift toward control predictions rather than symmetric noise.

All four interventions changed at least some gauge-aware diagnostics, but those changes did not track predictive effects closely. Across the 72 site-density cells, the absolute descriptive Spearman correlations between diagnostic changes and balanced-accuracy changes were at most 0.21 (figure 8). The measured representation summaries therefore do not provide a simple predictive mechanism or a causal mediation result.

4.5 Site, seed, and weighting sensitivity

Only 1 of the 18 leave-one-site influence estimates was positive, and no favorable exclusion interval lay above zero. Two of five seed point estimates favored BuNN, but no favorable seed interval excluded zero; one unfavorable interval lay below zero. Participant weighting reversed the sign of the small BuNN–GCN point estimate, whereas the equal-site mean and median

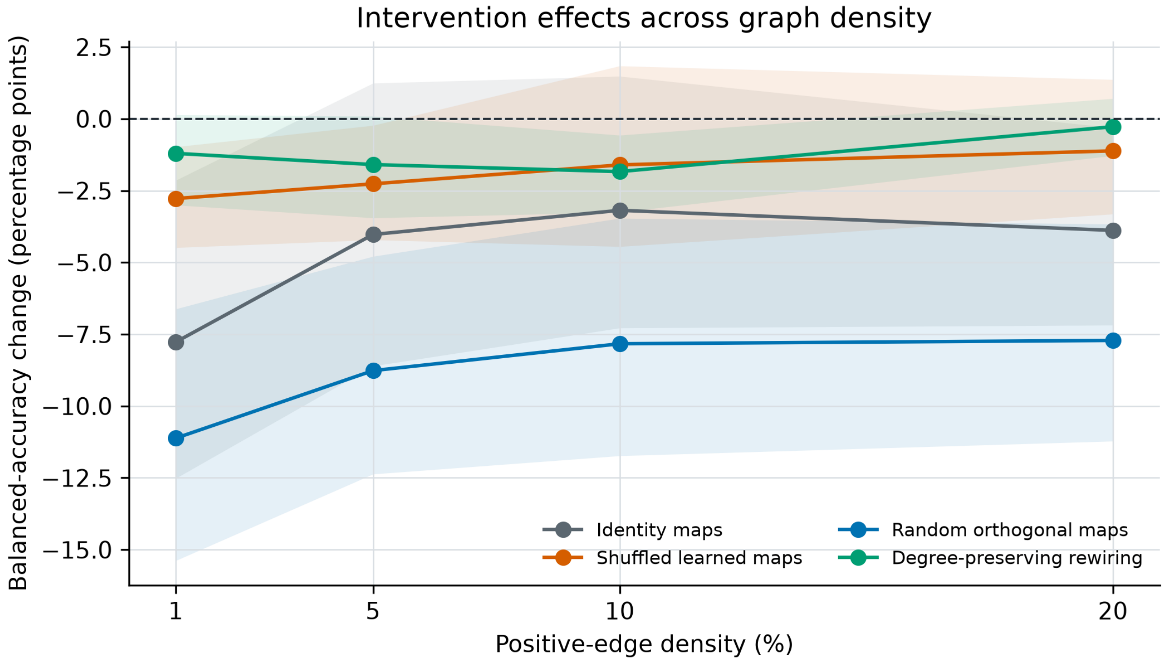

Figure 7: Density-specific accepted-checkpoint intervention effects. Shaded regions are 95% paired site-bootstrap intervals. These density-specific estimates are descriptive; the primary inference averages the four effects within site before equal-site analysis.

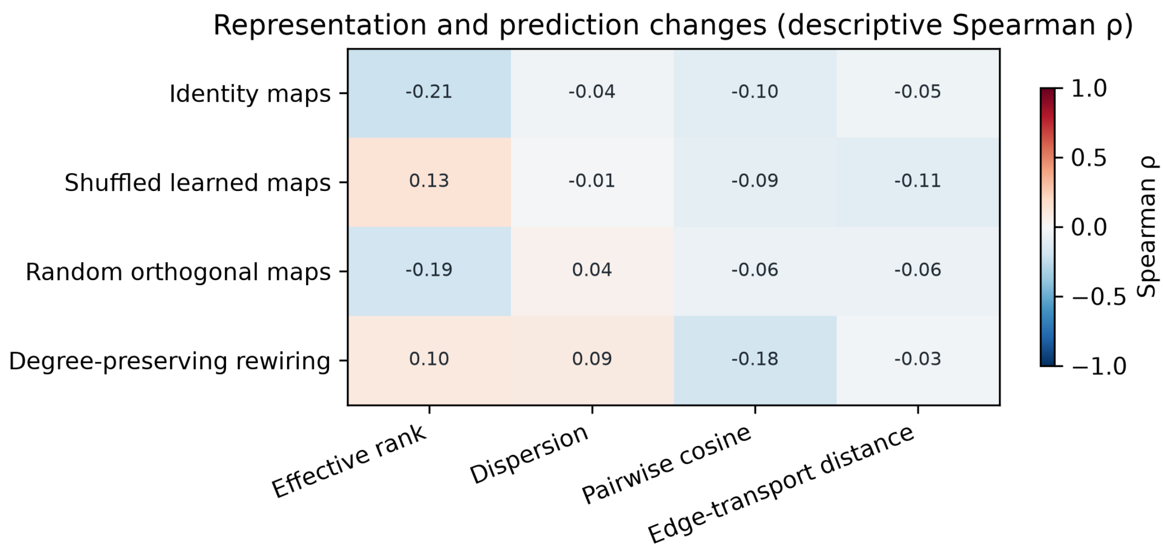

Figure 8: Descriptive association between final-layer gauge-aware diagnostic changes and balanced-accuracy changes across 72 site-density cells. The correlations are exploratory summaries, not multiplicity-controlled tests or evidence of causal mediation.

remained negative. The pre-specified confirmatory estimand was the equal-site mean.

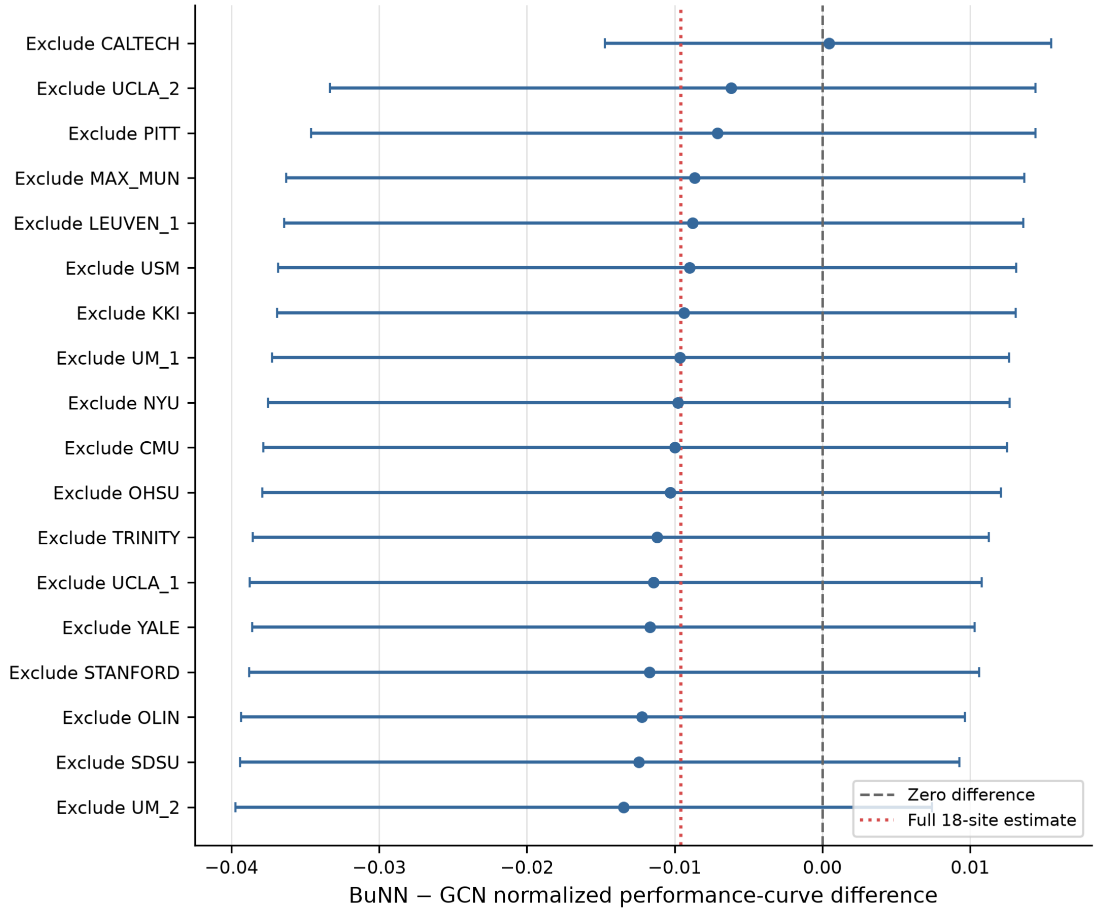

Figure 9: Leave-one-site influence on the BuNN–GCN curve contrast. This analysis tests whether the confirmatory average is driven by a single site; it does not create 18 independent replications.

4.6 Execution cost

Learned BuNN used 8,529 parameters, compared with 5,889 for GCN. Across the recorded fits, BuNN required 31.14 total fit-hours and GCN required 15.09. Maximum recorded peak GPU memory remained below 0.38 GiB for both. The low memory use reflects small graphs and models; the workload was dominated by many independent fits rather than a single large tensor. Runtime figures are specific to this implementation and the A10G system.

Table 7: Observed operator cost and equal-site curve performance.

Operator	Parameters	Fit-hours	Peak GPU GiB	Curve BA
GCN	5,889	15.09	0.373	0.5945
Trivial bundle	5,889	25.99	0.377	0.5911
Learned BuNN	8,529	31.14	0.378	0.5849

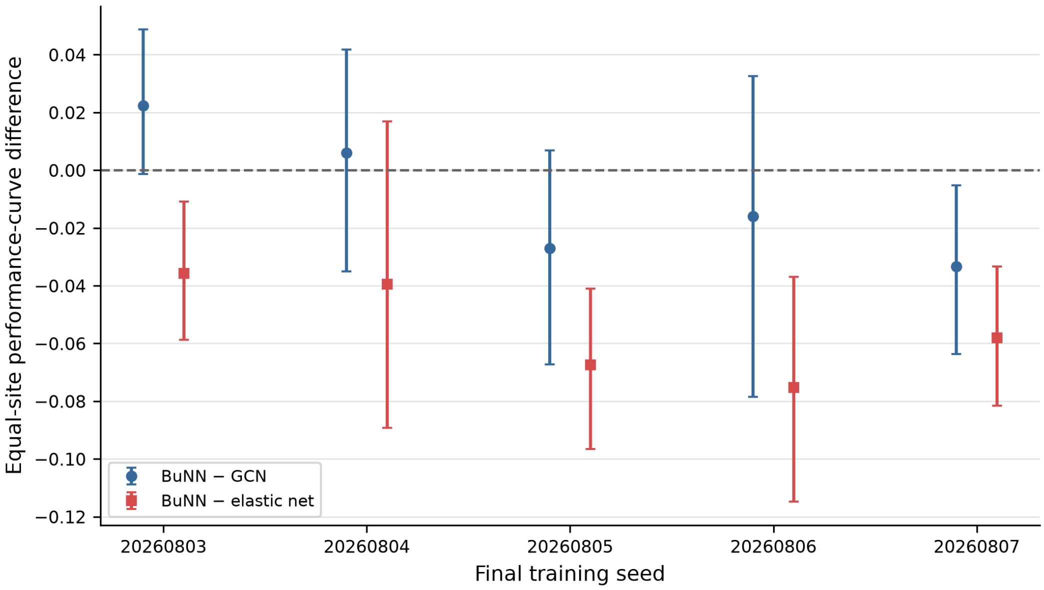

Figure 10: BuNN–GCN contrast across the five fixed final training seeds. Variation across seeds shows that the small average difference is sensitive to optimization randomness.

5 Discussion

The proposed BuNN advantage did not carry into this ABIDE-I pipeline. Learned bundle transport did not improve the performance–density curve over GCN, did not beat the strongest regularized connectome baseline, and did not preserve matched-anchor effective rank more strongly. Because inputs, backbone, tuning budget, and operator controls were held fixed, the result speaks directly to this propagation design rather than to a broad comparison of unrelated models.

The intervention audit resolves one ambiguity in that negative comparison. BuNN's learned maps were not ignored: replacing them with random orthogonal maps caused a large, multiplicity-controlled loss of held-out performance. The failure to beat GCN or the elastic net therefore cannot be explained by saying that training never used the map system. Instead, computationally important learned transformations existed inside a model whose overall inductive bias still did not improve cross-site prediction.

5.1 Why the result was plausible before the experiment

The original hypothesis was not arbitrary. Ordinary diffusion can contract node representations, while orthogonal transport can align features before averaging. In graph problems where neighboring features are expressed in different useful frames, that operation can protect structure that direct averaging would erase. BuNN theory and prior synthetic evaluations therefore provided a reasonable architectural motivation [2].

The experiment nevertheless placed that idea in an unusually demanding setting. Functional connectomes are dense, and the node feature is already a global profile. The benefit of transporting locally available information may be small when every node already describes its relationship with all other nodes.

5.2 What the completed evidence can explain

The data support a layered explanation, not a single proven mechanism.

Table 8: Evidence-based interpretation of the null result.

Observation	Supported interpretation	What it does not prove
Diffusion curves decline with density	Adding more propagation edges is associated with worse held-out-site prediction in this pipeline.	A universal law of over-smoothing or a causal biological mechanism.
BuNN does not beat GCN	Learned transport provides no detected predictive advantage over ordinary aggregation here.	That BuNN never works on brain or non-brain graphs.
BuNN remains below elastic net	A regularized model on the full connectome is the stronger practical reference for these inputs.	That nonlinear modeling is always unnecessary.
Learned-local anchor already has high raw rank	Extra node-wise map capacity can create an apparent raw diversity advantage without transport.	That every aspect of the learned maps is useless.
Matched-anchor rank favors GCN	The proposed common-frame representation-preservation advantage is not supported.	A causal mediation pathway from rank to prediction.
Random maps reduce BA by 8.86 points	Accepted predictions depend on properties of the learned map system.	Biological bundle geometry or causal neural transport.
Map shuffling has a smaller effect	The learned map collection matters more than the exact node assignment under this intervention.	That node identity is irrelevant in every architecture.
Degree-preserving rewiring reduces BA by 1.23 points	The constructed propagation graph contributes modestly to prediction.	Anatomical validity of the graph or a universal topology effect.
Diagnostic–performance correlations are weak	Large representation changes do not reliably co-vary with predictive changes across site-density cells.	Proof that representations can never mediate prediction.
Site and seed sensitivity	The small BuNN–GCN difference is not stable to all reasonable weighting and optimization choices.	That the full result is invalid or that site effects explain every difference.

The strongest current explanation is therefore architectural and data-based. The full connectivity rows already expose global information, while denser propagation averages those profiles. Learned orthogonal maps materially shape BuNN's own predictions, but the much smaller topology-rewiring effect suggests that message passing over the selected positive-edge graph contributes less. The model can learn useful internal coordinate transformations without those transformations outperforming a simpler GCN or a linear model that directly uses the complete connectome. This interpretation fits both Study 1 and E1; it is not a causal proof of which internal feature components were lost.

5.3 Why the learned-local control matters

BuNN introduces additional parameters through its map generator. A comparison with GCN alone could credit transport for differences caused by that added capacity. The learned-local control keeps the map generator but removes inter-node diffusion. Its high zero-density effective rank shows how a naive raw-embedding analysis could overstate preservation. Anchoring each operator to its corresponding no-propagation state makes the mechanism comparison more credible.

5.4 Site heterogeneity and uncertainty

The primary BuNN–GCN confidence interval crossed zero, and participant weighting reversed the point-estimate sign. Individual site exclusions and training seeds changed the magnitude and sometimes the direction of this small difference. The correct conclusion is not that the models are exactly equivalent. It is that no BuNN advantage was detected and that a gain of 0.03 balanced accuracy on the pre-specified curve estimand is inconsistent with the observed upper confidence bound.

The contrast against the connectome elastic net is more stable. Its interval remained below zero, indicating that the learned BuNN curve underperformed the regularized non-graph reference in this analysis. That is an important result because a paper comparing only neural graph variants could miss the strongest available baseline.

5.5 Limitations

The conclusions are conditional on ABIDE-I, C-PAC filtered data without global-signal regression, the AAL-116 atlas, Pearson correlations and Fisher transformation, positive-edge propagation graphs, full connectivity-row node features, the 0–20 percent density grid, the shared two-layer backbone, and the fixed tuning budget. Other atlases, preprocessing pipelines, graph construction rules, node features, or BuNN architectures may behave differently.

The study uses observational functional associations. It cannot identify anatomy, excitation, inhibition, or causal neural communication. The technical quality-control rule does not remove demographic or clinical sampling bias. Site sample sizes differ, and an equal-site mean answers a different question from a participant-weighted mean. The representation diagnostics are computational summaries; their association with prediction is not a causal mediation analysis.

E1 was specified only after the Study 1 result was known, so it is a post-hoc mechanistic audit rather than an independent confirmation. Its interventions are deliberately artificial. Random maps can disrupt the coordinate system expected by downstream learned weights, identity maps remove both learned orientation and reflections, and node shuffling preserves the learned map distribution while breaking its assignment. Their effect sizes reveal functional dependence under these checkpoints, not the effect of changing a single isolated biological mechanism. Finally, the completed report does not contain an external ABIDE-II validation cohort.

5.6 Remaining extensions and their evidential role

The accepted-checkpoint intervention audit is now complete and independently audited. It remains separate from the confirmatory result because its protocol was frozen after Study 1 was known. Deeper matched GCN/BuNN networks, a leakage-free Wang-style time-series model, synthetic ground-truth geometry, and prospective ABIDE-II evaluation remain separate studies. They should not be blended into either completed analysis or described as results until their protocols and outputs are frozen and audited.

6 Conclusion

In this pre-specified ABIDE-I analysis, learned orthogonal bundle transport did not improve the held-out-site performance–density curve over GCN. It also failed to surpass identity propagation or the learned-local control and remained below the regularized connectome baseline. Common-frame, matched-anchor diagnostics did not support the proposed representation-preservation advantage.

The finding does not show that bundle geometry is biologically false or that BuNNs are generally ineffective. It shows something narrower and useful: for dense functional-connectome graphs whose node features already contain global connectivity information, this learned transport operator added capacity and runtime without a detected advantage over GCN or elastic net. The post-hoc checkpoint audit adds an important qualification. Learned maps were computationally active: replacing them with random orthogonal maps reduced balanced accuracy by 8.86 points after multiplicity-controlled inference. Exact node-map placement and degree-preserving topology contributed less, and representation changes were only weakly associated with prediction changes.

The combined evidence therefore explains the negative comparison without dismissing BuNN as inert. It learned consequential internal transformations, but those transformations did not convert the selected graph and globally informed features into better cross-site classification. That distinction is the main contribution of the completed study. It was obtained through matched controls, held-out-site evaluation, strong non-graph baselines, gauge-aware diagnostics, frozen decision rules, checkpoint interventions, and independent artifact audits.

Evidence Boundary and Remaining Work

Table 9: Boundary between completed evidence and planned work.

Component	Completed and audited	Not yet a result
Data	ABIDE-I C-PAC filtered, no-GSR AAL time series; technical QC; 754-person connectome archive.	Alternative preprocessing, atlases, negative-edge propagation, ABIDE-II.
Classical models	Covariate, connectome elastic-net, and combined baselines across all 18 held-out sites.	Additional harmonization or feature-selection families.
Neural models	Two-layer identity, learned-local, GCN, trivial-bundle, and learned-BuNN operator study; five seeds; complete density grid.	Five-layer matched study, Wang-inspired time-series architecture.
Mechanism evidence	Common-frame representation curves, matched anchors, site influence, seed stability, efficiency, and the audited 360-checkpoint map/topology intervention study.	Component-level feature-loss analysis and ground-truth synthetic geometry.
External validity	None claimed beyond the frozen ABIDE-I pipeline.	Prospective preprocessing-compatible ABIDE-II evaluation.

A Reproducibility Record

The repository separates immutable scientific artifacts from presentation files. Raw participant derivatives, checkpoints, and credentials are excluded from the public release package. The report draws its numbers from generated tables and macros tied to accepted analysis hashes. The following checks were completed before the Study 1 result was interpreted:

- all 18 outer sites were present;
- expected tuning, prediction, metric, diagnostic, and runtime row counts were complete;
- model and data contracts matched their frozen hashes;
- hidden metrics were independently recomputed before unblinding;
- parallel worker outputs matched the validated sequential execution in the engineering equivalence test;
- site-bootstrap and sign-flip analyses were built and tested on synthetic data before use on accepted results;
- report figures and tables were regenerated from the frozen result snapshot rather than edited manually.

The current source code, report source, and manifests are stored in the project repository. Participant-level source data are not embedded in this PDF.

References

[1] K. Han, Y. Su, L. He, L. Zhan, S. Plis, V. Calhoun, and C. Yang, "Rethinking functional brain connectome analysis: Do graph deep learning models help?" *npj Artificial Intelligence*, vol. 2, p. 19, 2026. [Online]. Available: https://doi.org/10.1038/s44387-025-00067-x

[2] J. Bamberger, F. Barbero, X. Dong, and M. M. Bronstein, "Bundle neural network for message diffusion on graphs," in *International Conference on Learning Representations*, 2025. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/2ac879d1865475a7abc8dfc7a9c15c27-Abstract-Conference.html

[3] C. Bodnar, F. Di Giovanni, B. P. Chamberlain, P. Li'o, and M. M. Bronstein, "Neural sheaf diffusion: A topological perspective on heterophily and oversmoothing in GNNs," in *Advances in Neural Information Processing Systems*, vol. 35, 2022. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2022/hash/75c45fca2aa416ada062b26cc4fb7641-Abstract.html

[4] K. Murphy and M. D. Fox, "Towards a consensus regarding global signal regression for resting state functional connectivity MRI," *NeuroImage*, vol. 154, pp. 169–173, 2017. [Online]. Available: https://doi.org/10.1016/j.neuroimage.2016.11.052

[5] A. Di Martino *et al.*, "The autism brain imaging data exchange: Towards a large-scale evaluation of the intrinsic brain architecture in autism," *Molecular Psychiatry*, vol. 19, no. 6, pp. 659–667, 2014. [Online]. Available: https://doi.org/10.1038/mp.2013.78

[6] Preprocessed Connectomes Project, "ABIDE preprocessed," INDI, Child Mind Institute, 2026, accessed 9 August 2026. [Online]. Available: https://fcon_1000.projects.nitrc.org/indi/abide/preprocessed.html

[7] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *International Conference on Learning Representations*, 2017. [Online]. Available: https://openreview.net/forum?id=SJU4ayYgl

[8] N. Tzourio-Mazoyer, B. Landeau, D. Papathanassiou, F. Crivello, O. Etard, N. Delcroix, B. Mazoyer, and M. Joliot, "Automated anatomical labeling of activations in SPM using a macroscopic anatomical parcellation of the MNI MRI single-subject brain," *NeuroImage*, vol. 15, no. 1, pp. 273–289, 2002. [Online]. Available: https://doi.org/10.1006/nimg.2001.0978

[9] H. Zou and T. Hastie, "Regularization and variable selection via the elastic net," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 67, no. 2, pp. 301–320, 2005. [Online]. Available: https://doi.org/10.1111/j.1467-9868.2005.00503.x